LiPo Battery



Category: **Drones & Aviation** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: LiPo Battery; Lipo Battery and Charger

What it is used for

A lithium-polymer (LiPo) battery supplies electrical energy to drones and other portable equipment and is commonly charged using a compatible balance charger. Correct voltage selection, balanced charging, careful handling, and appropriate storage are essential because damaged or incorrectly charged LiPo batteries can ignite or release hazardous fumes.

Main parts

- Battery cells store electrical energy in a lithium-based chemistry.
- Battery pack casing or wrapping protects the cell assembly.
- Main power connector supplies energy to the equipment.
- Balance connector provides access to individual cell voltages.
- Compatible charger controls the charging process.
- Charger display or indicators show voltage, current, and charge status.
- Battery management or protection circuit is included where fitted.
- Fire-resistant charging surface or approved charging container supports safer charging.

Before you start

1. Only trained users should charge and handle LiPo batteries, following the battery and charger instructions.
2. Check the battery chemistry, cell count, rated voltage, capacity, and permitted charge rate.
3. Inspect the battery for swelling, punctures, crushed areas, damaged insulation, loose wires, or unusual odour.
4. Do not charge a damaged, swollen, wet, or abnormally hot battery.
5. Use a charger with the correct LiPo mode and compatible cell-count and balance functions.
6. Place the battery on a non-combustible surface in a well-ventilated area away from combustible materials.
7. Check the charger, leads, connectors, and balance plug for damage and correct polarity.

How to use

1. Confirm the battery and charger specifications match before connecting anything.
2. Connect the charger to its power source according to the charger instructions.
3. Connect the battery's main lead and balance connector to the correct charger ports without forcing the connectors.
4. Select the correct LiPo chemistry, cell count, and approved charging settings.
5. Verify the displayed cell count and voltage before starting the charge.

6. Start charging and remain nearby, observing the battery and charger throughout the process.
7. Stop charging immediately if the battery swells, becomes unusually hot, leaks, smells abnormal, or the charger reports a fault.
8. When charging is complete, stop the charger and disconnect the battery using the approved sequence.
9. Allow the battery to cool to a suitable temperature before use or storage.
10. For storage, use the manufacturer's recommended storage voltage and a suitable cool, dry location.
11. Record battery condition, charging issues, and any battery removed from service.

⚠ Safety

- Never leave a LiPo battery charging unattended.
- Never use a damaged or swollen battery or attempt to puncture, compress, or repair it yourself.
- Do not charge beyond the battery manufacturer's voltage, current, or temperature limits.
- Prevent short circuits by keeping metal objects away from battery terminals and exposed connectors.
- Do not charge near flammable materials, escape routes, or in direct sunlight or excessive heat.
- If a battery smokes, hisses, swells rapidly, or ignites, move away, warn others, and follow the site's battery-fire emergency procedure; do not handle a failing battery.

Care & storage

- Inspect battery casing, wires, connectors, and balance leads before and after use.
- Keep terminals clean and dry and protect connectors from impact and contamination.
- Store batteries at the recommended storage voltage in a cool, dry, fire-conscious location.
- Separate damaged batteries from serviceable stock and follow approved hazardous-battery disposal procedures.
- Use only compatible chargers and arrange replacement when capacity, condition, or safety becomes unacceptable.

Common mistakes to avoid

- Selecting the wrong cell count or battery chemistry on the charger.
- Charging a swollen, punctured, or otherwise damaged battery.
- Leaving the battery charging without supervision.
- Storing a LiPo battery fully charged or fully discharged for an extended period against manufacturer guidance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.